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KTG & THERMODYNAMICS_MICRO_XI



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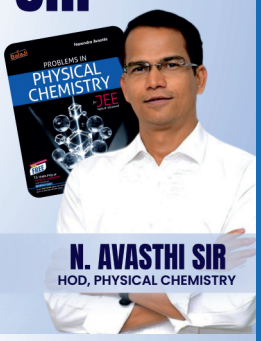

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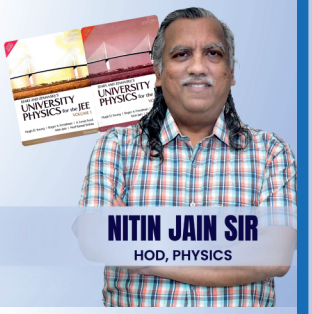


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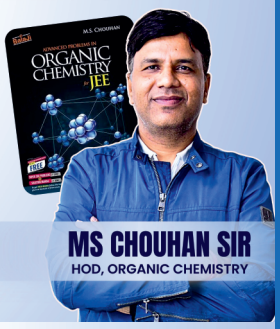


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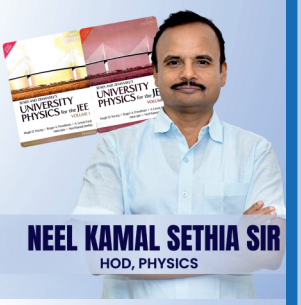


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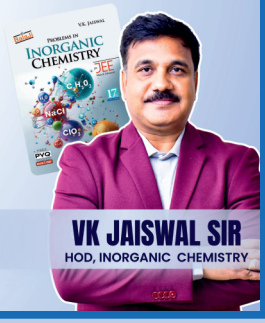


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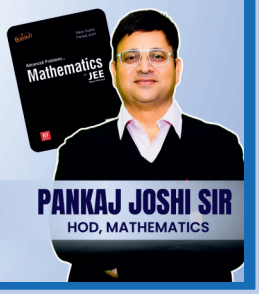


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PHYSICS

TARGET IIT JEE
MICRO-XI

KINETIC THEORY OF GASES
&
THERMODYNAMICS

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KEY CONCEPTS

Kinetic Theory Of Gases

1. Assumption of kinetic theory of gases

(1) A gas consist of particles called molecules which move randomly in all directions.
(2) These molecules obey Newton's law of motion.
(3) Size of molecule negligible in comparison to average separation between the molecules.
(4) The forces on molecule are negligible except at the time of collision.
(5) All collision between molecules or between molecules and wall are pefectly elastic. Time of collision is very small.
(6) For large number of molecules the density and distribution of molecules with different velocities are independent of position, direction and time.

2. Pressure of an ideal gas

$P = \frac{1}{3}\rho\bar{v}^2 = \frac{1}{3}\rho\bar{v}^2_{rms}$ Here $\bar{v}$ = mean square speed

v_{rms} = root mean square speed
ρ = density of gas

$P = \frac{2}{3}(\frac{1}{2}\rho v^2_{rms})$

$P = \frac{2}{3}E$

$E = \frac{3}{2}P$

So total K.E. $K = \frac{3}{2}PV2$

3. R.M.S. velocity – depends on tempearture only for any gas.

$$V_{rms} = \sqrt{\frac{v_1^2+v_2^2+v_3^2+.....+v_n^2}{n}}$$

$P = \frac{1}{3}\rho v^2_{rms}$

$v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M}}$

4. Most Probable velocity – velocity which maximum number of molecules may have

$v_{mp} = \sqrt{\frac{2RT}{M}}$

5. Average velocity

$v_{avg} = \frac{\bar{v}_1+\bar{v}_2+.....+\bar{v}_n.}{n} = 0$

6. Average speed

$$v_{avg} = \frac{|\vec{v}_1| + |\vec{v}_2| + |\vec{v}_3| + ...+|\vec{v}_n|.}{n} = \sqrt{\frac{8RT}{\pi M}}$$

7. Ideal gas equation

PV= nRT (container form of gas law/ pressure volume form)

$P = \left(\frac{\rho}{M}\right)RT$ (open atmosphere / pressure density form)

8. Graham's law of diffusion :–

When two gases at the same pressure and tempearture are allowed to diffuse into each other the rate of diffusion of each gas is inversely proportional to the square root of the density of the gas

$r \alpha v_{rms}$ where r = rate of diffusion

so, $\frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}}$

9. Degree of Freedom (f) – No. of ways in which a gas molecule can distribute its energy

10. Law of equipartition of energy : – Energy in each degree of freedom = 1/2 KT joules

If degree of freedom is f. Energy = $\frac{f}{2}$ KT joules.

$$U = \frac{f}{2} KTnN_A = \frac{f}{2} nRT$$

11. Degree of freedom(f) in different gas molecules

Molecules	Translational	Rotational
Monoatomic	3	0
Diatomic	3	2
Polyatomic	3	2 (linear molecule) 3 (non-linear molecule)

Translational energy for all type of molecules = $\frac{3}{2}$ **(nRT)**

Law of Thermodynamics

1. Zeroth law of thermodynamics :– If two bodies A and B are in thermal equilibrium and A and C are also in thermal equilibrium.Then B and C are also in thermal equilibrium.

2. First law of Thermodynamics:– Energy conservation for gaseous system.

Heat supplied to the gas = Increment in internal energy + work done by the gas.

$\Delta Q = \Delta U + \Delta W$ ΔQ is +ve for heat supplied

in differential form $dQ = dU + dW$ ΔQ is –ve for heat rejected

and $dQ = nCdT$ C = molar specific heat

$C = C_P$ (constant pressure) ; $C = C_V$ = (constant volume)

$$dU = \frac{f}{2} nRdT$$

$$dW = \int_{v_1}^{v_2} P\,dv$$ (P = pressure of the gas of which work is to be calculated)

ΔW = +ve for work done by gas (in expansion of gas)

ΔW = –ve for work done on the gas (in contraction of gas)

Molar specific heat for a given process $C = \frac{f}{2} R + \frac{R\,PdV}{PdV + VdP} = C_V + \frac{R\,PdV}{PdV + VdP}$

Process	C	Monoatomic	Diatomic	Polyatomic
V= constant	$C_V = (f/2)R$	(3/2)R	(5/2)R	3R
P = constant	$C_P = \frac{f+2}{2}R$	(5/2)R	(7/2)R	4R

Mayor's Relation $C_P = C_V + R$

Note :– C of a gas depends on the process of that gas, which can be infinite in types.

Ratio of specific heat :– $\gamma = \frac{C_P}{C_V} = \frac{f+2}{f}$

- Monoatomic → 5/3 = 1.67
- Diatomic → 7/5 = 1.4
- Polyatomic → 4/3 = 1.33

and $f = \frac{2}{\gamma - 1}$

$C_V = \frac{R}{\gamma - 1}$; $C_P = \frac{\gamma R}{\gamma - 1}$

Isochoric Process (V= constant)

$dV = 0 \Rightarrow dW = 0$

By FLT $dQ = dU = nC_V\,dT$

$$Q = \int_{T_1}^{T_2} nC_v\,dT = nC_v(T_2 - T_1)$$

* Be careful if $\Delta V = 0$ then not necessarily an Isochoric Process.

Isothermal Process (T = constant)

$dT = 0$, $dU = 0$

$$Q = W = (nRT) \int_{v_1}^{v_2} dV/V$$

Isobaric Process (P = constant)

$dP = 0$

By FLT $dQ = dU + dW$

$$n_{Cp}(T_2 - T_1) = \left(\frac{f}{2}\right)nR(T_2 - T_1) + nR(T_2 - T_1)$$

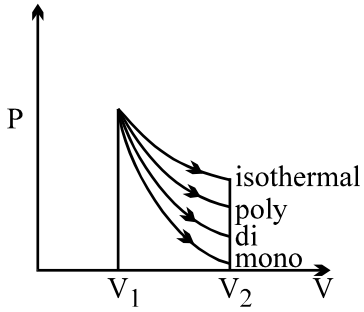
$$W = nR(T_2 - T_1)$$

* If $\Delta P = 0$ then not necessarily an Isobaric Process.

$$W = nRT\ ln\frac{V_2}{V_1} = nRT\ ln\frac{P_1}{P_2} \qquad \left(\frac{V_2}{V_1} = \frac{P_1}{P_2} = \text{compression ratio}\right)$$

Adiabatic Process dQ = 0 but if $\Delta Q = 0$, it is not necessaserily adibatic.
dW = –dU By FLT

$$W = \int_{T_1}^{T_2} \frac{nRdT}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{P_1V_1 - P_2V_2}{\gamma - 1}$$



So $PdV + VdP = (\gamma - 1)$ .....(ii)
For Adiabatic Process PV^{γ} = constant
Work done is least for monoatomic gas

$$\left|\frac{dP}{dV}\right|_{adiabatic} = \gamma\left|\frac{dP}{dV}\right|_{isothermal}$$

Polytropic process

PV^n= constant

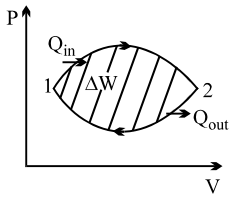
$$P = \frac{K}{V^n} \Rightarrow \frac{dP}{dV} = n\left|\frac{K}{V^{n+1}}\right| \quad ; \qquad C = \frac{R}{\gamma - 1} + \frac{R}{1 - n}$$

So C is constant for polytropic process

Efficiency of a cyclic process

$\Delta U = 0$
so $\Delta Q = \Delta W$

Efficiency $\eta = \dfrac{\text{work done by gas}}{\text{heat input}}$



$$\eta = \frac{W}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

EXERCISE-I

1. Find the approx. number of molecules contained in a vessel of volume 7 litres at 0°C at 1.3×10^5 pascal
(A) 2.4×10^{23} (B) 3×10^{23} (C) 6×10^{23} (D) 4.8×10^{23}

2. An ideal gas mixture filled inside a balloon expands according to the relation $PV^{2/3}$ = constant. The temperature inside the balloon is
(A) increasing (B) decreasing (C) constant (D) can't be said

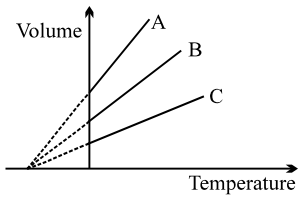
3. At a temperature T K, the pressure of 4.0g argon in a bulb is p. The bulb is put in a bath having temperature higher by 50K than the first one. 0.8g of argon gas had to be removed to maintained original pressure. The temperature T is equal to
(A) 510 K (B) 200 K (C) 100 K (D) 73 K

4. A vessel of volume 0.02 m^3 contains a mixture of hydrogen and helium at 20°C and 2 atmospheric pressure. The mass of mixture is 5 gms. Find the ratio of mass of hydrogen to that of helium in the mixture.
(A) 1 : 2 (B) 1 : 3 (C) 2 : 3 (D) 3 : 2

5. During an experiment an ideal gas obeys an addition equation of state P^2V = constant. The initial temperature and volume of gas are T and V respectively. When it expands to volume 2V, then its temperature will be :
(A) T (B) $\sqrt{2}$ T (C) 2 T (D) $2\sqrt{2}$ T

6. The temperature of a gas is doubled (i) on absolute scale (ii) on centigrade scale. The increase in root mean square velocity of gas will be
(A) More in case (i) (B) More in case (ii)
(C) Same in both case (D) Information not sufficient

7. The expansion of an ideal gas of mass m at a constant pressure P is given by the straight line B. Then the expansion of the same ideal gas of mass 2 m at a pressure 2P is given by the straight line



(A) C (B) A (C) B (D) none

8. A container X has volume double that of contianer Y and both are connected by a thin tube. Both contains same ideal gas. The temperature of X is 200K and that of Y is 400K. If mass of gas in X is m then in Y it will be:
(A) m/8 (B) m/6 (C) m/4 (D) m/2

9. One mole of an ideal gas at STP is heated in an insulated closed container until the average speed of its molecules is doubled. Its pressure would therefore increase by factor.
(A) 1.5 (B) $\sqrt{2}$ (C) 2 (D) 4

10. The ratio of specific heats of a gas is 9/7, then the number of degrees of freedom of the gas molecules for translational motion is :

(A) 7 (B) 3 (C) 6 (D) none

11. An ideal gas undergoes an adiabatic process obeying the relation $PV^{4/3}$ = constant. If its initial temperature is 300 K and then its pressure is increased upto four times its initial value, then the final temperature is (in Kelvin):

(A) $300\sqrt{2}$ (B) $300\sqrt[3]{2}$ (C) 600 (D) 1200

12. One mole of an ideal gas at temperature T_1 expends according to the law $\frac{P}{V^2}$ = a (constant). The work done by the gas till temperature of gas becomes T_2 is :

(A) $\frac{1}{2}R(T_2 - T_1)$ (B) $\frac{1}{3}R(T_2 - T_1)$ (C) $\frac{1}{4}R(T_2 - T_1)$ (D) $\frac{1}{5}R(T_2 - T_1)$

13. Why does the pressure of ideal gas increase when it is heated at constant volume ?

(A) The gas molecules expand

(B) The molecules move at the same speed, but hit the walls more often

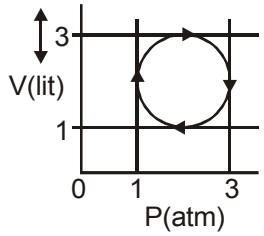
(C) The molecules move faster and hit the walls more often

(D) The number of molecules of gas increases

14. The temperature of n moles of an ideal gas is increasd from T to 4T through a process for which pressure $p = aT^{-1}$ where a is a constant. Then, the work done by the gas is (R is universal gas constant)

(A) nRT (B) 4nRT (C) 2nRT (D) 6nRT

15. Net heat energy absorbed by the system in going through a cyclic process shown in the figure is (1 atm = 10^5 Pa)



(A) $-10^2\pi$J (B) πJ (C) $10^2\pi$J (D) $-\pi$J

16. A system is taken from a given initial state to a given final state along various paths represented on a p-V diagram. The quantity that is independent of the path is :

(A) amount of heat transferred Q (B) amount of work done W

(C) (Q – W) (D) None of these

17. In thermodynamic process pressure of a fixed mass of gas is changed in such a manner that the gas releases 30 joule of heat and 18 joule of work was done on the gas. If the initial internal energy of the gas was 60 joule, then, the final internal energy will be :

(A) 32 joule (B) 48 joule (C) 72 joule (D) 96 joule

18. 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :

(A) 32 kJ (B) 5 kJ (C) – 4 kJ (D) 13 kJ

19. A vessel contains an ideal monoatomic gas which expands at constant pressure, when heat Q is given to it. Then the work done in expansion is :

(A) Q (B) $\frac{3}{5}Q$ (C) $\frac{2}{5}Q$ (D) $\frac{2}{3}Q$

20. One mole of an ideal monoatomic gas at temperature T_0 expands slowly according to the law P/V = constant. If the final temperature is $2\,T_0$, heat supplied to the gas is :

(A) $2\,R\,T_0$ (B) $\frac{3}{2}\,R\,T_0$ (C) $R\,T_0$ (D) $\frac{1}{2}\,R\,T_0$

21. Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the masses of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively. The temperature of the mixture will be :

(A) $\frac{T_1+T_2}{n_1+n_2}$ (B) $\frac{T_1}{n_1}+\frac{T_2}{n_2}$ (C) $\frac{n_2T_1+n_1T_2}{n_1+n_2}$ (D) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$

Question No. 22 to 25 (4 questions)

Five moles of helium are mixed with two moles of hydrogen to form a mixture. Take molar mass of helium M_1 = 4g and that of hydrogen M_2 = 2g

22. The equivalent molar mass of the mixture is

(A) 6g (B) $\frac{13g}{7}$ (C) $\frac{18g}{7}$ (D) none

23. The equivalent degree of freedom f of the mixture is

(A) 3.57 (B) 1.14 (C) 4.4 (D) none

24. The equivalent value of γ is

(A) 1.59 (B) 1.53 (C) 1.56 (D) none

25. If the internal energy of He sample of 100J and that of the hydrogen sample is 200J, then the internal energy of the mixture is

(A) 900 J (B) 128.5 J (C) 171.4 J (D) 300 J

26. N(<100) molecules of a gas have velocities1, 2, 3..... N/km/s respectively. Then

(A) rms speed and average speed of molecules is same.

(B) ratio of rms speed to average speed is $\sqrt{(2N+1)(N+1)/6N}$

(C) ratio of rms speed to average speed is $\sqrt{(2N+1)(N+1)/6}$

(D) ratio of rms speed to average speed of a molecule is $2/\sqrt{6} \times \sqrt{(2N+1)/(N+1)}$

27. A reversible adiabatic path on a P-V diagram for an ideal gas passes through state A where $P = 0.7 \times 10^5$ N/m^{-2} and v = 0.0049 m^3. The ratio of specific heat of the gas is 1.4. The slope of path at A is

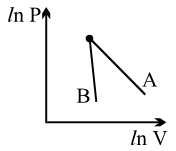
(A) 2.0×10^7 Nm^{-5} (B) 1.0×10^7 Nm^{-5}

(C) -2.0×10^7 Nm^{-5} (D) -1.0×10^7 Nm^{-5}

28. One mole of an ideal gas is contained with in a cylinder by a frictionless piston and is initially at temperature T. The pressure of the gas is kept constant while it is heated and its volume doubles. If R is molar gas constant, the work done by the gas in increasing its volume is

(A) RT *l*n2 (B) 1/2 RT (C) RT (D) 3/2 RT

29. The figure, shows the graph of logarithmic reading of pressure and volume for two ideal gases A and B undergoing adiabatic process. From figure it can be concluded that



(A) gas B is diatomic (B) gas A and B both are diatomic

(C) gas A is monoatomic (D) gas B is monoatomic & gas A is diatomic

30. A thermodynamic cycle takes in heat energy at a high temperature and rejects energy at a lower temperature. If the amount of energy rejected at the low temperature is 3 times the amount of work done by the cycle, the efficiency of the cycle is

(A) 0.25 (B) 0.33 (C) 0.67 (D) 0.9

31. Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume becomes half the initial volume.

(A) If the compression is adiabatic then monoatomic gas will have maximum final pressure.

(B) If the compression is adiabatic then triatomic gas will have maximum final pressure.

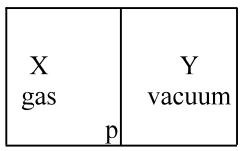
(C) If the compression is adiabatic then their final pressure will be same.

(D) If the compression is isothermal then their final pressure will be different.

32. A polyatomic gas with six degrees of freedom does 25J of work when it is expanded at constant pressure. The heat given to the gas is

(A) 100J (B) 150J (C) 200J (D) 250J

33. A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y. Which of the following statements is correct?



(A) No work is done by X (B) X decreases in temperature

(C) X increases in internal energy (D) X doubles in pressure

34. The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is

(A) 0 Nm^{-2} (B) 1 Nm^{-2} (C) 1.4×10^4 Nm^{-2} (D) 1.4×10^5 Nm^{-2}

35. If the molecules in a tank of hydrogen have the same RMS speed as the molecules in another tank of oxygen, we may be sure that :

(A) the pressures are the same

(B) the hydrogen is at the higher temperature

(C) the temperatures are the same

(D) the oxygen is at the higher temperature

36. Certain perfect gas is found to obey $PV^{3/2}$ = constant during an adiabatic process. If such a gas at initial temperature T is adiabatically compressed to half the initial volume, its final temperature will be

(A) $\sqrt{2}$ T (B) 2T (C) $2\sqrt{2}$ T (D) 4T

37. At very high temperatures vibrational degrees also becomes active. At such temperatures an ideal diatomic gas has a molar specific heat at constant pressure, C_P =

(A) 3R/2 (B) 5R/2 (C) 6R/2 (D) 9R/2

38. For a monatomic ideal gas undergoing an adiabatic change, the relation between temperature and volume is TV^x = constant where x is :

(A) $\frac{7}{5}$ (B) $\frac{2}{5}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

39. A perfect gas of a given mass is heated first in a small vessel and then in a large vessel, such that their volumes remain unchanged. The P-T curves are

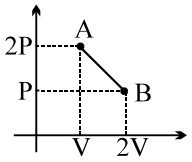
(A) parabolic with same curvature

(B) parabolic with different curvature

(C) linear with same slopes

(D) linear with different slopes

40. The process AB is shown in the diagram. As the gas is taken from A to B, its temperature



(A) initially increases then decreases

(B) initially decreases then increases

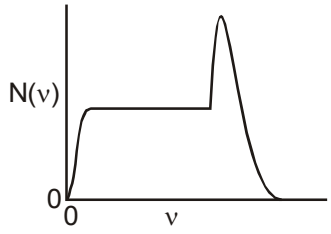
(C) remains constant

(D) variation depends on type of gas

41. At temperature T, N molecules of gas A each having mass m and at the same temperature 2N molecules of gas B each having mass 2m are filled in a container. The mean sqaure velocity of molecules of gas B is v^2 and mean square of x component of velocity of molecules of gas A is w^2. The ratio of w^2/v^2 is :

(A) 1 (B) 2 (C) 1/3 (D) 2/3

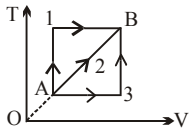
42. Consider the distribution of speeds of molecules as shown in figure. Which is the correct ordering for the speeds?



(A) $\nu_{rms} < \nu_{av} < \nu_p$ (B) $\nu_{rms} < \nu_p < \nu_{av}$ (C) $\nu_{av} < \nu_{rms} < \nu_p$ (D) $\nu_{av} < \nu_p < \nu_{rms}$

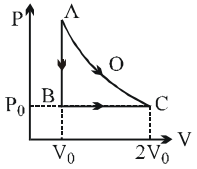
43. Temperature of a body is only a manifestation of the mean
(A) total mechanical energy of a molecule of the body
(B) potential energy of a molecule of the body
(C) rotational kinetic energy of a molecule of the body
(D) translational kinetic energy of a molecule of the body

44. A given mass of a gas expands from a state A to the state B by three paths 1, 2 and 3 as shown in T-V indicator diagram. If W_1, W_2 and W_3 respectively be the work done by the gas along the three paths, then



(A) $W_1 > W_2 > W_3$ (B) $W_1 < W_2 < W_3$ (C) $W_1 = W_2 = W_3$ (D) $W_1 < W_2$, $W_1 > W_3$

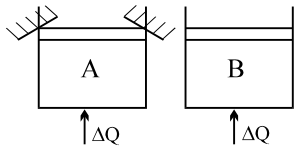
45. An ideal gas is taken from point A to point C on P-V diagram through two process AOC and ABC as shown in the figure. Process AOC is isothermal



(A) Process AOC requires more heat than process ABC.
(B) Process ABC requires more heat than process AOC.
(C) Both process AOC & ABC require same amount of heat.
(D) Data is insufficient for comparison of heat requirement for the two processes.

46. A cylinder made of perfectly non conducting material closed at both ends is divided into two equal parts by a heat proof piston. Both parts of the cylinder contain the same masses of a gas at a temperature $t_0 = 27°$ and pressure $P_0 = 1$ atm. Now if the gas in one of the parts is slowly heated to t = 57°C while the temperature of first part is maintained at t_0 the distance moved by the piston from the middle of the cylinder will be (length of the cylinder = 84 cm)
(A) 3 cm (B) 5 cm (C) 2 cm (D) 1 cm

47. Two identical vessels A & B contain equal amount of ideal monoatomic gas. The piston of A is fixed but that of B is free. Same amount of heat is absorbed by A & B. If B's internal energy increases by 100 J the change in internal energy of A is



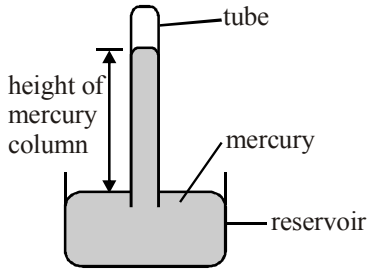
(A) 100 J (B) $\frac{500}{3}$ J (C) 250 J (D) none of these

48. Two moles of monoatomic gas is expanded from (P_0, V_0) to $(P_0, 2V_0)$ under isobaric condition. Let ΔQ_1, be the heat given to the gas, ΔW_1 the work done by the gas and ΔU_1 the change in internal energy. Now the monoatomic gas is replaced by a diatomic gas. Other conditions remaining the same. The corresponding values in this case are ΔQ_2 , ΔW_2 , ΔU_2 respectively, then

(A) $\Delta Q_1 - \Delta Q_2 = \Delta U_1 - \Delta U_2$ (B) $\Delta U_2 + \Delta W_2 > \Delta U_1 + \Delta W_1$

(C) $\Delta U_2 > \Delta U_1$ (D) All of these

49. The diagram shows a simple mercury barometer. Which of the following does not cause the height of the mercury column to vary ?



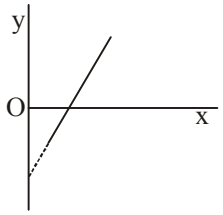
(A) changes in the atmospheric pressure

(B) changes in the value of g

(C) evaporation of mercury from the barometer reservoir

(D) leakage of air into the tube

50. The expansion of a perfect gas of mass 0.01 kg at constant pressure is shown in the figure where x and y respectively represent

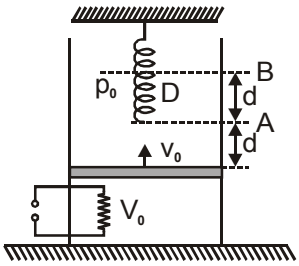


(A) temperature (°C) and volume (B) temperature (K) and volume

(C) volume and temperature (°C) (D) volume and temperature (K)

Paragraph for question nos. 51 to 53

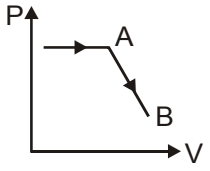
A mass less piston, which can move without friction, closes a sample of Helium in a vertical, thermally insulated cylinder, which is closed at its bottom, and the cross-section of which is A = 2 dm^2. Above the piston there is a fixed stand to which an unstretched spring of spring constant 2000 N/m is attached, whose bottom end is at a distance of 2 dm from the piston when the volume of the gas is V_0 = 8 dm^3. The external pressure is $p_0 = 10^5$ Pa, g = 10 m/s^2. The gas confined in the cylinder is heated with an electric heating element. (1 dm = 10^{-1} m)

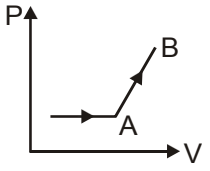


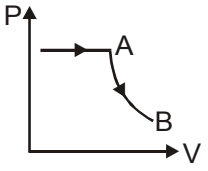
51. How much heat is supplied by the element till it reaches the spring :

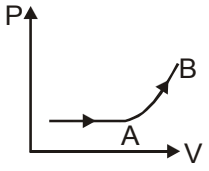
(A) Zero (B) 600 J (C) 8000 J (D) 1000 J

52. How will process look like on a PV diagram

(A) 

(B) 

(C) 

(D) 

53. What is the ratio of work done by gas in moving the piston from initial position to A, and from there to B

(A) $\frac{10}{11}$ (B) $\frac{11}{10}$ (C) $\frac{5}{3}$ (D) $\frac{3}{5}$

55. An ideal gas expands from volume V_1 to V_2. This may be achieved by either of the three processes: isobaric, isothermal and adiabatic. Let ΔU be the change in internal energy of the gas, Q be the quantity of heat added to the system and W be the work done by the system on the gas. Identify which of the following statements is false for ΔU?

(A) ΔU is least under adiabatic process.

(B) ΔU is greatest under adiabatic process.

(C) ΔU is greatest under the isobaric process.

(D) ΔU in isothermal process lies in-between the values obtained under isobaric and adiabatic processes.

56. **Statement-1 :** Temperature of Real gases often changes when undergoing a free expansion under adiabetic condition whereas that of an ideal gas does not.

Statement-2 : First law of thermodynamics holds for ideal goes only.

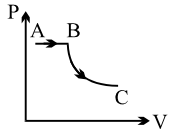
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

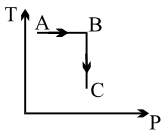
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

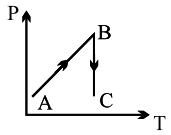
(C) Statement-1 is true, statement-2 is false.

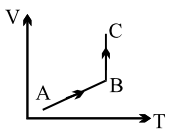
(D) Statement-1 is false, statement-2 is true.

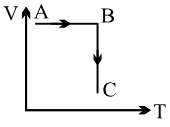
57. A process is shown in the diagram. Which of the following curves may represent the same process?



(A) 

(B) 

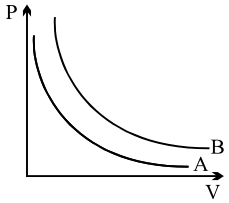
(C) 

(D) 

58. An ideal gas expands in such a way that PV^2 = constant throughout the process.
(A) The graph of the process of T-V diagram is a parabola.
(B) The graph of the process of T-V diagram is a straight line.
(C) Such an expansion is possible only with heating.
(D) Such an expansion is possible only with cooling.

59. A gas expands such that its initial and final temperature are equal. Also, the process followed by the gas traces a straight line on the P-V diagram :
(A) The temperature of the gas remains constant throughout.
(B) The temperature of the gas first increases and then decreases
(C) The temperature of the gas first decreases and then increases
(D) The stright line has a negative slope.

60. Figure shows the pressure P versus volume V graphs for two different gas sample at a given temperature. M_A and M_B are masses of two samples, n_A and n_B are numbers of moles. Which of the following ***must be incorect.***



(A) $M_A > M_B$ (B) $M_A < M_B$ (C) $n_A > n_B$ (D) $n_A < n_B$

61. During an experiment, an ideal gas is found to obey a condition VP^2 = constant. The gas is initially at a temperature T, pressure P and volume V. The gas expands to volume 4V.
(A) The pressure of gas changes to P/2
(B) The temperature of gas changes to 4T
(C) The graph of above process on the P-T diagram is parabola
(D) The graph of above process on the P-T diagram is hyperbola.

62. According to kinetic theory of gases,
(A) The velocity of molecules decreases for each collision
(B) The pressure exerted by a diatomic gas is proportional to the mean velocity of the molecule.
(C) The K.E. of the gas decreases on expansion at constant temperature.
(D) The mean translational K.E. of a diatomic gas increases with increase in absolute temperature.

63. A piston is slowly pushed into a metal cylinder containing an ideal gas. Which of the following statements is/are incorrect?
(A) The pressure of the gas increases
(B) The number of the molecules per unit volume increases
(C) The average speed of gas molecules increases
(D) The frequency of collision of the gas molecules with the piston increases.

64. Hydrogen gas and oxygen gas have volume 1 cm^3 each at N.T.P.
(A) Number of molecules is same in both the gases.
(B) The rms velocity of molecules of both the gases is the same.
(C) The internal energy of each gases is the same
(D) The average velocity of molecules of each gas is the same.

65. A student records ΔQ, ΔU & ΔW for a thermodynamic cycle $A \to B \to C \to A$. Certain entries are missing. Find correct entry in following options.

	AB	BC	CA
ΔW	40J		30J
ΔU		50J	
ΔQ	150J	10J	

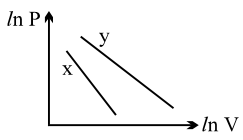
(A) $W_{BC} = -70$ J (B) $\Delta Q_{CA} = 130$ J (C) $\Delta U_{AB} = 190$ J (D) $\Delta U_{CA} = -160$ J

66. Two gases have the same initial pressure, volume and temperatue. They expand to the same final volume, one adiabatically and the other isothermally
(A) The final temperature is greater for the isothermal process
(B) The final pressure is greater for the isothermal process
(C) The work done by the gas is greater for the isothermal process
(D) All the above options are incorrect

67. The first law of thermodynamics can be written as $\Delta U = \Delta Q + \Delta W$ for an ideal gas. Which of the following statements is correct?
(A) ΔU is always zero when no heat enters or leaves the gas
(B) ΔW is the work done by the gas in this written law.
(C) ΔU is zero when heat is supplied and the temperature stays constant
(D) $\Delta Q = -\Delta W$ when the temperature increases very slowly.

68. 2 moles of a monoatomic gas are expanded to double its initial volume, through a process P/V = constant. If its initial temperature is 300 K, then which of the following is not true.
(A) $\Delta T = 900$ K (B) $\Delta Q = 3200$ R (C) $\Delta Q = 3600$ R (D) W = 900 R

69. For two different gases X and Y, having degrees of freedom f_1 and f_2 and molar heat capacities at constant volume C_{V_1} and C_{V_2} respectively, the *l*n P versus *l*n V graph is plotted for adiabatic process, as shown

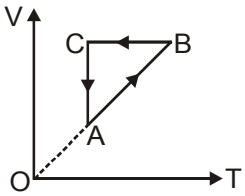


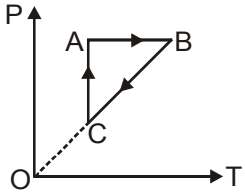
(A) $f_1 > f_2$ (B) $f_2 > f_1$ (C) $C_{V_2} > C_{V_1}$ (D) $C_{V_1} > C_{V_2}$

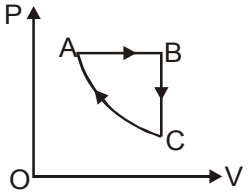
70. An ideal gas is subjected to a thermodynamic process in which its pressure increases but volume decreases. Then
(A) the process may be isothermal
(B) work done by the gas is positive
(C) the process may be adiabatic
(D) change in its internal energy is equal to the work done by the gas

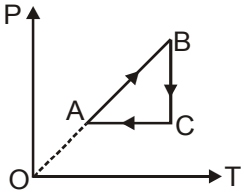
71. An ideal gas confined into an insulated chamber is allowed to enter into an evacuated insulated chamber. If Q, W and ΔE_{int} have the usual meanings, then
(A) $Q = 0, W \neq 0$ (B) $W = 0, Q \neq 0$ (C) $\Delta E_{int} = 0, Q \neq 0$ (D) $Q = W = \Delta E_{int} = 0$

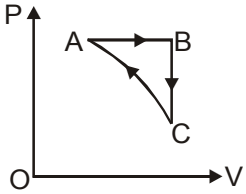
72. An ideal gas undergoes a thermodynamics cycle as shown in figure. Which of the following graphs represents the same cycle?



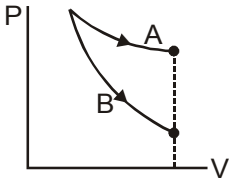
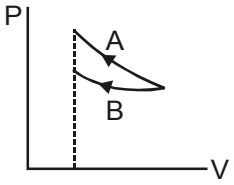
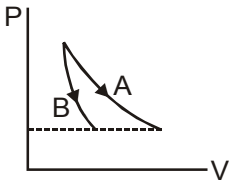
(*A) 

(B) 

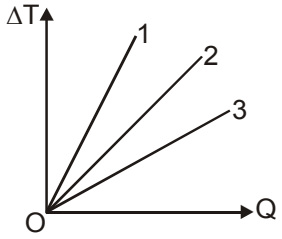
(C) 

(D) 

73. An ideal gas undergoes two processes A and B. One of these is isothermal and the other is adiabatic.

Column-I		**Column-II**	
(A)		(P)	Heat supplied during curve A is positive
(B)		(Q)	Work done by gas in both processes positive
(C)		(R)	Internal energy increases in adiabatic process
		(S)	Temperature of gas in pocess B is constant

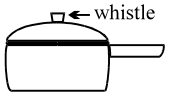
74. The straight lines in the figure depict the variations in temperature ΔT as a function of the amount of heat supplied Q in different process involving the change of schange of state of a monoatomic and a distomic ideal gas. The initial states (P, V, T) of the two gases are the same. Fill in the blanks correct number of the straight line as shown in the figure.



(A) Isobaric process of monoatomic gas.....	(P)	1
(B) Isobaric process of diatomic gas.....	(Q)	2
(C) isochoric process of monoatomic gas.....	(R)	3
(D) Isochoric process of diatomic gas.....	(S)	x-axis (i.e. 'Q' axis)

EXERCISE-II

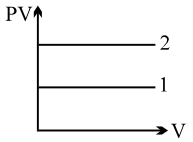
1. An empty pressure cooker of volume 10 litres contains air at atmospheric pressure 10^5 Pa and temperature of 27°C. It contains a whistle which has area of 0.1 cm^2 and weight of 100 gm. What should be the temperature of air inside so that the whistle is just lifted up?

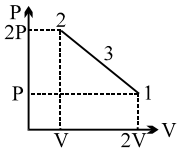


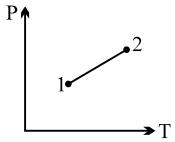
2. Air at temperature of 400 K and atmospheric pressure is filled in a balloon of volume 1 m^3. If surrounding air is at temperature of 300 K, find the ratio of Buoyant force on balloon and weight of air inside

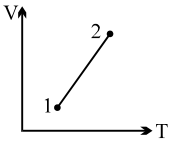
3. A closed vessel of volume V_0 contains oxygen at a pressure P_0 and temperature T_0. Another closed vessel of the same volume V_0 contains helium at a pressure of P_0 and temperature $T_0/2$. Find the ratio of the masses of oxygen to the helium.

4. Examine the following plots and predict whether in (i) $P_1 < P_2$ and $T_1 > T_2$, in (ii) $T_1 = T_2 < T_3$, in (iii) $V_1 > V_2$, in (iv) $P_1 > P_2$ or otherwise.

(i)


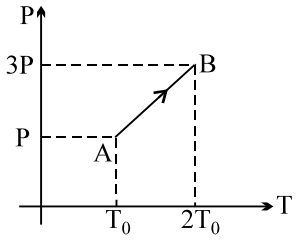
(ii)


(iii)


(iv)


5. A piston divides a closed gas cylinder into two parts. Initially the piston is kept pressed such that one part has a pressure P and volume 5V and the other part has pressure 8P and volume V. The piston is now left free. Find the new pressures and volumes for the adiabatic and isothermal processes. For this gas $\gamma = 1.5$.

6. Pressure versus temperature graph of an ideal gas is shown. Density of gas at point A is ρ_0. Find the density of gas at B.



7. The average degrees of freedom per molecules for a gas is 6. The gas performs 25 J of work when it expands at constant pressure. Find the heat absorbed by the gas .

8. A mixture of 4 gm helium and 28 gm of nitrogen in enclosed in a vessel of constant volume 300°K. Find the quantity of heat absorbed by the mixture to doubled the root mean square velocity of its molecules.
(R = Universal gas constant)

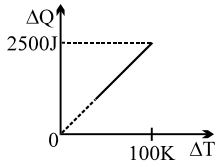
9. One mole of an ideal gas is compressed from 0.5 lit to 0.25 lit. During the compression, 23.04×10^2 J of work is done on the gas and heat is removed to keep the temperature of the gas constant at all times. Find the temperature of the gas. (Take universal gas constant R = 8.31 J $mol^{-1}K^{-1}$)

10. If heat is added at constant volume, 6300 J of heat are required to raise the temperature of an ideal gas by 150 K. If instead, heat is added at constant pressure, 8800 joules are required for the same temperature change. When the temperature of the gas changes by 300 K. Determine the change is the internal energy of the gas.

11. 70 calorie of heat is required to raise the temperature of 2 mole of an ideal gas at constant pressure from 40°C to 45°C. Find the amount of heat required to raise the temperature of the same gas through the same range at constant volume (R = 2 cal/mol-K)

12. The volume of one mole of an ideal gas with specific heat ratio γ is varied according to the law $V = \frac{a}{T^2}$, where a is a constant. Find the amount of heat obtained by the gas in this process if the gas temperature is increased by ΔT.

13. One mole of a gas mixture is heated under constant pressure, and heat required ΔQ is plotted against temperature difference acquired. Find the value of γ for mixture.



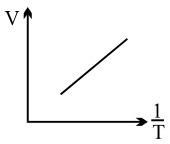
14. A gas is undergoing an adiabatic process. At a certain stage A, the values of volume and temperature $\equiv (V_0, T_0)$ and the magnitude of the slope of V-T curve is m. Find the value of C_P and C_V.

15. 1 mole of an ideal gas at initial temperature T was cooled isochorically till the gas pressure decreased n times. Then by an isobaric process, the gas was restored to the initial temperature T. Find the net heat absorbed by the gas in the whole process.

16. A gas undergoes a process in which the pressure and volume are related by VP^n = constant. Find the bulk modulus of the gas.

17. Ideal diatomic gas is taken through a process $\Delta Q = 2\Delta U$. Find the molar heat capacity for the process (where ΔQ is the heat supplied and ΔU is change in internal energy)

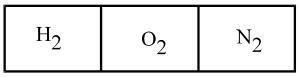
18. One mole of an ideal monoatomic gas undergoes a process as shown in the figure. Find the molar specific heat of the gas in the process.



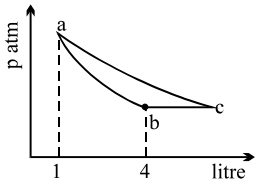
19. The pressure of an ideal gas changes with volumes as P = aV where 'a' is a constant. One moles of this gas is expanded to 3 times its original volume V_0. Find
(i) the heat transferred in the process. (ii) the heat capacity of the gas.

20. A barometer is faulty. When the true barometer reading are 73 and 75 cm of Hg, the faulty barometer reads 69 cm and 70 cm respectively.
(i) What is the total length of the barometer tube?
(ii) What is the true reading when the faulty barometer reads 69.5 cm ?
(iii) What is the faulty barometer reading when the true barometer reads 74 cm?

21. A vessel of volume V = 30/ is separated into three equal parts by stationary semipermeable thin membranes as shown in the Figure. The left , middle and right parts are filled with m_{H_2} = 30g of hydorgen, m_{O_2} =160g of oxygen, and m_{N_2} = 70g of nitrogen respectively. The left partition lets through only hydrogen, while the right partition lets through hydrogen and nitrogen. What will be the pressure in each part of the vessel after the equilibrium has been set in if the vessel is kept at a constant temperature T = 300K?

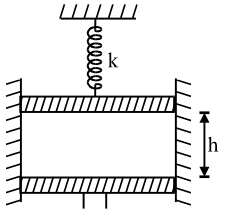


22. Figure shows three processes for an ideal gas. The temperature at 'a' is 600 K, pressure 16 atm and volume 1 litre. The volume at 'b' is 4 litre. Out of the two process ab and ac, one is adiabatic and the other is isothermal. The ratio of specific heats of the gas is 1.5. Answer the following :

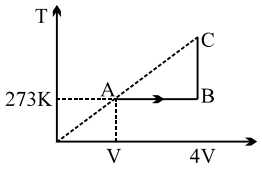


(i) Which of ab and ac processes is adiabatic. Why?
(ii) Compute the pressure of the gas at b and c.
(iii) Compute the temperature at b and c.
(iv) Compute the volume at c.

23. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section A = 27 cm^2, between two light movable pistons as shown in the figure. Spring with force constant k = 3700 N/m is in a relaxed state initially. Now the lower piston is moved upwards a height h/2, h being the initial length of gas column. It is observed that the upper piston moves up by a distance h/16. Find h taking γ for the gas to be 1.5. Also find the final temperature of the gas.

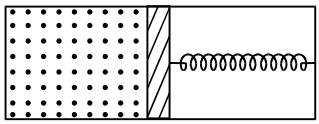


24. At a temperature of $T_0 = 273°K$, two moles of an ideal gas undergoes a process as shown. The total amount of heat imparted to the gas equals Q = 27.7 kJ. Determine the ratio of molar specific heat capacities.

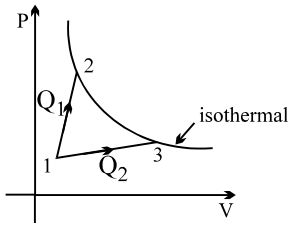


25. A parallel beam of particles of mass m moving with velocities v impinges on a wall at an angle θ to its normal. The number of particles per unit volume in the beam is n. If the collision of particles with the wall is elastic, then find the pressure exerted by this beam on the wall.

26. A thermally insulated vessel is divided into two parts by a heat-insulating piston which can move in the vessel without the friction. The left part of the vessel contains one mole of an ideal monatomic gas, & the right part is vacuum. The piston is connected to the right wall of the vessel through a spring whose length in free state is equal to the length of the vessel as shown in the figure. Determine the heat capacity C of the system, neglecting the heat capacities of the vessel, piston and spring.



27. A gas takes part in two processes in which it is heated from the same initial state 1 to the same final temperature. The processes are shown on the P-V diagram by the straight line 1-2 and 1-3. 2 and 3 are the points on the same isothermal curve. Q_1 and Q_2 are the heat transfer along the two processes. Then in which case will the heat transfer be more.



EXERCISE-III

1. A container of fixed volume has a mixture of one mole of hydrogen and one mole of helium in equilibrium at temperature T. Assuming the gases are ideal, the correct statement(s) is(are) **[JEE Advanced 2015]**

(A) The average energy per mole of the gas mixture is 2RT.

(B) The ratio of speed of sound in the gas mixture to that in helium gas is $\sqrt{6/5}$.

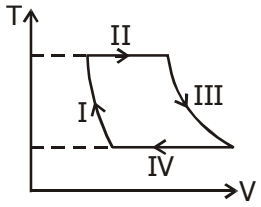
(C) The ratio of the rms speed of helium atoms to that of hydrogen molecules is 1/2.

(D) The ratio of the rms speed of helium atoms to that of hydrogen molecules is $1/\sqrt{2}$

2. A gas is enclosed in a cylinder with a movable frictionless piston. Its initial thermodynamic state at pressure $P_i = 10^5$Pa and volume $V_i = 10^{-3}$ m^3 changes to a final state at $P_f = (1/32) \times 10^5$ Pa and $V_f = 8 \times 10^{-3}$ m^3 in an adiabatic quasi-static process, such that P^3V^5 = constant. Consider another thermodynamic process that brings the system from the same initial state to the same final state in two steps : an isobaric expansion at P_i followed by an isochoric (isovolumetric) process at volume V_f. The amount of heat supplied to the system in the two-step process is approximately. **[JEE Advance 2016]**

(A) 112 J (B) 294 J (C) 588 J (D) 813 J

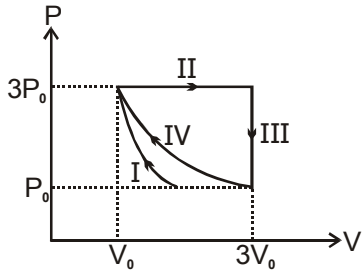
3. One mole of a monatomic ideal gas undergoes a cyclic process as shown in the figure (where V is the volume and T is the temperature). Which of the statements below is(are) true? **[JEE Advanced 2018]**



(A) Process I is an isochoric process (B) In process II, gas absorbs heat

(C) In process IV, gas releases heat (D) Processes I and III are **not** isobaric

4. One mole of a monatomic ideal gas undergoes four thermodynamic processes as shown schematically in the PV-diagram below. Among these four processes, one is isobaric, one is isochoric, one is isothermal and one is adiabatic. Match the processes mentioned in List-I with the corresponding statements in List-II. **[JEE Advanced 2018]**



	List-I		**List-II**
(P)	In process I	(1)	Work done by the gas is zero
(Q)	In process II	(2)	Temperature of the gas remains unchanged
(R)	In process III	(3)	No heat is exchanged between the gas and its surroundings
(S)	In process IV	(4)	Work done by the gas is $6P_0V_0$

(A) P $\rightarrow$ 4 ; Q $\rightarrow$ 3 ; R $\rightarrow$ 1 ; S $\rightarrow$ 2 (B) P $\rightarrow$ 1 ; Q $\rightarrow$ 3 ; R $\rightarrow$ 2 ; S $\rightarrow$ 4

(C) P $\rightarrow$ 3 ; Q $\rightarrow$ 4 ; R $\rightarrow$ 1 ; S $\rightarrow$ 2 (D) P $\rightarrow$ 3 ; Q $\rightarrow$ 4 ; R $\rightarrow$ 2 ; S $\rightarrow$ 1

5. One mole of a monatomic ideal gas undergoes an adiabatic expansion in which its volume becomes eight times its initial value. If the initial temperature of the gas is 100 K and the universal gas constant R = 8.0 J $mol^{-1}K^{-1}$, the decrease in its internal energy, in Joule, is _____. **[JEE Advanced 2018]**

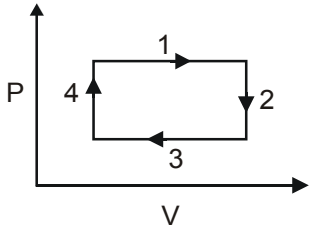
6. A mixture of ideal gas containing 5 moles of monatomic gas and 1 mole of rigid diatomic gas is initially at pressure P_0, volume V_0, and temeprature T_0. If the gas mixture is adiabatically compressed to a volume $V_0/4$, then the correct statement(s) is/are,
(Given : $2^{1.2} = 2.3$; $2^{3.2} = 9.2$; R is gas constant) **[JEE Advanced 2019]**
(A) The average kinetic energy fo the gas mixture after compression is in between $18RT_0$ and $19RT_0$
(B) Adiabatic constant of the gas mixture is 1.6
(C) The work |W| done during the process is $13RT_0$
(D) The final pressure of the gas mixture after compression is in between $9P_0$ and $10P_0$

7. The rms speeds of the molecules of Hydrogen, Oxygen and Carbon dioxide at the same temperature are V_H, V_O and V_C respectively then : **[JEE Main Online 2021]**
(A) $V_H = V_O > V_C$ (B) $V_H = V_O = V_C$
(C) $V_H > V_O > V_C$ (D) $V_C > V_O > V_H$

8. An ideal gas is expanding such that PT^3 = constant. The coefficient of volume expansion of the gas is : **[JEE Main Online 2021]**
(A) 4/T (B) 3/T (C) 1/T (D) 2/T

9. A balloon carries a total load of 185 kg at normal pressure and temperature of 27°C. What load will the balloon carry on rising to a height at which the barometric pressure is 45 cm of Hg and the temperature is –7°C. Assuming the volume constant ? **[JEE Main Online 2021]**
(A) 181.46 kg (B) 123.54 kg (B) 219.07 kg (D) 214.15 kg

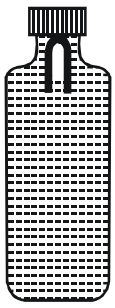
10. An ideal gas undergoes a four step cycle as shown in the P - V diagram below. During this cycle, heat is absorbed by the gas in **[JEE Advanced 2021]**



(A) steps 1 and 2 (B) steps 1 and 3 (C) steps 1 and 4 (D) steps 2 and 4

Question Stem for Question Nos. 11 and 12

A soft plastic bottle, filled with water of density 1 gm/cc, carries an inverted glass test-tube with some air (ideal gas) trapped as shown in the figure. The test-tube has a mass of 5 gm, and it is made of a thick glass of density 2.5 gm/cc. Initially the bottle is sealed at atmospheric pressure $P_0 = 10^5$ Pa so that the volume of the trapped air is $v_0 = 3.3$ cc. When the bottle is squeezed from outside at constant temperature, the pressure inside rises and the volume of the trapped air reduces. It is found that the test tube begins to sink at pressure $p_0 + \Delta p$ without changing its orientation. At this pressure, the volume of the trapped air is $v_0 - \Delta v$.
Let $\Delta v = X$ cc and $\Delta p = Y \times 10^3$ Pa. **[JEE Advanced 2021]**

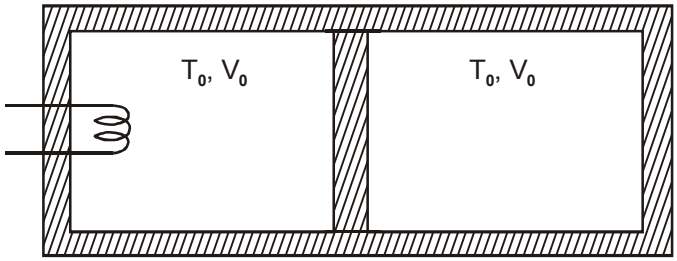


11. The value of X is ____.

12. The value of Y is ____.

Paragraph for Question Nos. 13 to 14

A thermally insulating cylinder has a thermally insulating and frictionless movable partition int he middle, as shown in the figure below. On each side of the partition, there is on mole of an ideal gas, with specific heat at constant volume, $C_V = 2R$. Here, R is the gas constant. Initially, each side has a volume V_0 and temperature T_0. The left side has an electric heater, which is turned on at very low power to transfer heat Q to the gas on the left side. As a result the partition moves slowly towards the right reducing the right side volume to $V_0/2$. Consequently, the gas temperatures on the left and the right sides become T_L and T_R, respectively. Ignore the changes in the temperatures of the cylinder, heater and the partition.



13. The value of $\frac{T_R}{T_0}$ is : **[JEE Advanced 2021]**

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) 3

14. The value of $\frac{Q}{RT_0}$ is :

(A) $4(2\sqrt{2}+1)$ (B) $4(2\sqrt{2}-1)$ (C) $(5\sqrt{2}+1)$ (D) $(5\sqrt{2}-1)$

15. What will be the effect on the root mean square velocity of oxygen molecules if the temperature is doubled and oxygen molecule dissociates into atomic oxygen? **[JEE Main Online 2022]**

(A) The velocity of atomic oxygen remains same
(B) The velocity of atomic oxygen doubles
(C) The velocity of atomic oxygen becomes half
(D) The velocity of atomic oxygen becomes four times

16. A mixture of hydrogen and oxygen has volume 2000 cm^3, temperature 300 K, pressure 100 kPa and mass 0.76 g. The ratio of number of moles of hydrogen to number of moles of oxygen in the mixture will be : [Take gas constant R = 8.3 JK^{-1} mol^{-1}] **[JEE Main Online 2022]**

(A) $\frac{1}{3}$ (B) $\frac{3}{1}$ (C) $\frac{1}{16}$ (D) $\frac{16}{1}$

17. A cylinder of fixed capacity of 44.8 litres contains helium gas at standard temperature and pressure. The amount of heat needed to raise the temperature of gas in the cylinder by 20.0°C will be : (Given gas constant R = 8.3 JK^{-1}-mol^{-1}) **[JEE Main Online 2022]**

(A) 249 J (B) 415 J (C) 498 J (D) 830 J

18. A Carnot engine whose heat sinks at 27ºC, has an efficiency of 25%. By how many degrees should the temperature of the source be changed to increase the efficiency by 100% of the original efficiency?

(A) Increases by 18ºC (B) Increases by 200ºC
(C) Increases by 120ºC (D) Increases by 73ºC **[JEE Main Online 2022]**

19. Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways. The work done by the gas is W_1 if the process is purely isothermal. W_2 if the process is purely adiabatic and W_3 if the process is purely isobaric. Then, choose the coned option **[JEE Main Online 2022]**

(A) $W_1 < W_2 < W_3$ (B) $W_2 < W_3 < W_1$ (C) $W_3 < W_1 < W_2$ (D) $W_2 < W_1 < W_3$

20. List-I describes thermodynamic processes in four different systems. List-II gives the magnitudes (either exactly or as a close approximation) of possible changes in the internal energy of the system due to the process. **[JEE Advanced 2022]**

List-I	**List-II**
(I) 10^{-3} kg of water at 100°C is converted to steam at the same temperature, at a pressure of 10^5 Pa. The volume of the system changes from 10^{-6} m^3 to 10^{-3} m^3 in the process. Latent heat of water = 2250 kJ/kg.	(P) 2 kJ
(II) 0.2 moles of a rigid diatomic ideal gas with volume V at temperature 500 K undergoes an isobaric expansion to volume 3 V. Assume R = 8.0 J mol^{-1} K^{-1}.	(Q) 7 kJ
(III) One mole of a monatomic ideal gas is compressed adiabatically from volume $V = \frac{1}{3} m^3$ and pressure 2 kPa to volume $\frac{V}{8}$.	(R) 4 kJ
(IV) Three moles of a diatomic ideal gas whose molecules can vibrate, is given 9 kJ of heat and undergoes isobaric expansion.	(S) 5 kJ (T) 3 kJ

Which one of the following options is correct?

(A) (I) → (T); (II) → (R); (III) → (S); (IV) → (Q) (B) (I) → (S); (II) → (P); (III) → (T); (IV) → (P)

(C) (I) → (P); (II) → (R); (III) → (T); (IV) → (Q) (D) (I) → (Q); (II) → (R); (III) → (S); (IV) → (T)

21. When a gas filled in a closed vessel is heated by raising the temperature by 1ºC, its pressure increases by 0.4%. The initial temperature of the gas is _____K. **[JEE Main Online 2022]**

22. The total internal energy of two mole monoatomic ideal gas at temperature T = 300 K will be J. (Given R = 8.31 J/mol.K) **[JEE Main Online 2022]**

23. 0.056 kg of Nitrogen is enclosed in a vessel at a temperature of 127ºC. The amount of heat required to double the speed of its molecules is _____ k cal. (Take : R = 2 cal mol^{-1} K^{-1}) **[JEE Main Online 2022]**

24. Match List-I with List-II : **[JEE Main Online 2023]**

List-I	**List-II**
(A) 3 Translation degrees of freedom	(I) Monoatomic gases
(B) 3 Translation, 2 rotational degrees of freedoms	(II) Polyatomic gases
(C) 3 Translation, 2 rotational and 1 vibrational degrees of freedom	(III) Rigid diatomic gases
(D) 3 Translation, 3 rotational and more than one vibrational degrees of freedom	(IV) Nonrigid diatomic gases

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(IV), (C)-(III), (D)-(II) (C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(C) (A)-(IV), (B)-(II), (C)-(I), (D)-(III) (D) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

25. Under isothermal condition, the pressure of a gas is given by $P = aV^{-3}$, where a is a constant and V is the volume of the gas. The bulk modulus at constant temperature is equal to **[JEE Main Online 2023]**

(A) P/2 (B) 2P (C) 3P (D) P

26. The mean free path of molecules of a certain gas at STP is 1500 d, where d is the diameter of the gas molecules. While maintaining the standard pressure, the mean free path of the molecules at 373 K is approximately. **[JEE Main Online 2023]**

(A) 1098 d (B) 1500 d (C) 2049 d (D) 750 d

27. If the r.m.s speed of chlorine molecule is 490 m/s at 27°C, the r.m.s speed of argon molecules at the same temperature will be
(Atomic mass of argon = 39.9 u, molecular mass of chlorine = 70.9 u) **[JEE Main Online 2023]**
(A) 751.7 m/s (B) 651.7 m/s (C) 451.7 m/s (D) 551.7 m/s

28. The temperature at which the kinetic energy of oxygen molecules becomes double than its value at 27°C is
(A) 1227°C (B) 627°C (C) 327°C (D) 927°C
[JEE Main Online 2023]

29. A gas is compressed adiabatically, which one of the follwing statement is NOT true.
[JEE Main Online 2023]
(A) The change in the internal energy is equal to the work done on the gas
(B) The temperature of the gas increases
(C) There is no heat supplied to the system
(D) There is no change in the internal energy

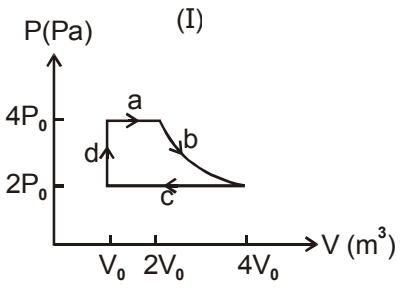
30. The Thermodynamics process, in which internal energy of the system remains constant is :
(A) isochoric (B) Isobaric (C) Adiabatic (D) Isothermal **[JEE Main 2023]**

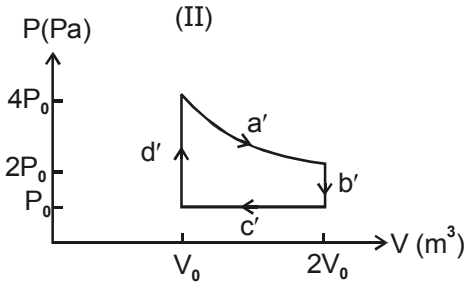
31. The initial pressure and volume of an ideal gas are P_0 and V_0. The final pressure of the gas when the gas is suddenly compressed to volume $\frac{V_0}{4}$ will be : (Given) γ = ratio of specific heats at constant pressure and at constant volume **[JEE Main Online 2023]**
(A) $P_0(4)P_0(4)^{\frac{1}{\gamma}}$ (B) $P_0(4)^{\gamma}$ (C) $4P_0$ (D) P_0

32. Work done by carnot engine operating between temperatures 127°C and 27°C is 2 kJ. The amount of heat transferred to the engine by the reservoir is : **[JEE Main Online 2023]**
(A) 2.27 kJ (B) 2 kJ (C) 8 kJ (D) 4 kJ

33. One mole of an ideal gas expands adiabatically from an initial state (T_A, V_0) to final state (T_f, $5V_0$). Another mole of the same gas expands isothermally from a different initial state (T_B, V_0) to the same final state (T_f, $5V_0$). The ratio of the specific heats at constant pressure and constant volume of this ideal gas is γ. What is the ratio T_A / T_B ? **[JEE Advanced 2023]**
(A) $5^{\gamma-1}$ (B) $5^{1-\gamma}$ (C) 5^{γ} (D) $5^{1+\gamma}$

34. One mole of an ideal gas undergoes two different cyclic processes I and II, as shown in the P-V diagrams below. In cycle I, processes a, b, c and d are isobaric, isothermal, isobaric and isochoric, respectively. In cycle II, processes a', b', c' and d' are isothermal, isochoric, isobaric and isochoric, respectively. The total work done during cycle I is W_I and that during cycle II is W_{II}. The ratio W_I / W_{II} is _____.
[JEE Advanced 2023]



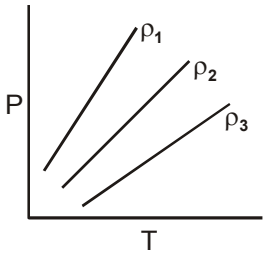


35. A closed container contains a homogeneous mixture of two moles of an ideal monatomic gas ($\gamma = 5/3$) and one mole of an ideal diatomic gas ($\gamma = 7/5$). Here, γ is the ratio of the specific heats at constant pressure and constant volume of an ideal gas. The gas mixture does a work of 66 Joule when heated at constant pressure. The change in its internal energy is _____ Joule. **[JEE Advanced 2023]**

36. A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is : **[JEE Main Online 2024]**

(A) $\frac{1}{4}$ (B) $\frac{1}{32}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{2\sqrt{2}}{1}$

37. P-T diagram of an ideal gas having three different densities ρ_1 , ρ_2 , ρ_3 (in three different cases) is shown in the figure. Which of the following is correct **[JEE Main Online 2024]**



(A) $\rho_1 < \rho_2$ (B) $\rho_1 = \rho_2 = \rho_3$ (C) $\rho_1 > \rho_2$ (D) $\rho_2 < \rho_3$

38. If the collision frequency of hydrogen molecules in a closed chamber at 27ºC is Z, then the collision frequency of the same system at 127ºC is : **[JEE Main Online 2024]**

(A) $\frac{\sqrt{3}}{2}Z$ (B) $\frac{2}{\sqrt{3}}Z$ (C) $\frac{3}{4}Z$ (D) $\frac{4}{3}Z$

39. If n is the number density and d is the diameter of the molecule, then the average distance covered by a molecule between two successive collisions (i.e. mean free path) is represented by : **[JEE Main Online 2024]**

(A) $\frac{1}{\sqrt{2n\pi d^2}}$ (B) $\frac{1}{\sqrt{2}n\pi d^2}$ (C) $\sqrt{2}n\pi d^2$ (D) $\frac{1}{\sqrt{2}\,n^2\pi^2 d^2}$

40. The translational degrees of freedom (f_t) and rotational degrees of freedom (f_r) of CH_4 molecule are : **[JEE Main Online 2024]**

(A) $f_t = 3$ and $f_r = 2$ (B) $f_t = 3$ and $f_r = 3$ (C) $f_t = 2$ and $f_r = 2$ (D) $f_t = 2$ and $f_r = 3$

41. The volume of an ideal gas ($\gamma = 1.5$) is changed adiabatically from 5 liters to 4 liters. The ratio of initial pressure to final pressure is : **[JEE Main Online 2024]**

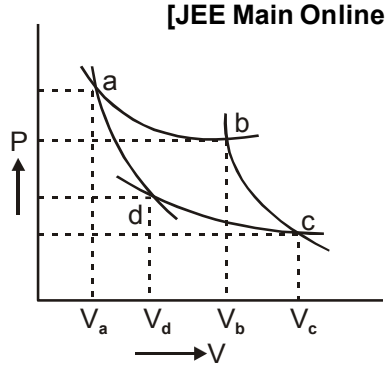
(A) $\frac{4}{5}$ (B) $\frac{8}{5\sqrt{5}}$ (C) $\frac{16}{25}$ (D) $\frac{2}{\sqrt{5}}$

42. A diatomic gas ($\gamma = 1.4$) does 100 J of work in an isobaric expansion. The heat given to the gas is : **[JEE Main Online 2024]**

(A) 350 J (B) 490 J (C) 150 J (D) 250 J

43. Two different adiabatic paths for the same gas intersect two isothermal curves as shown in P-V diagram.

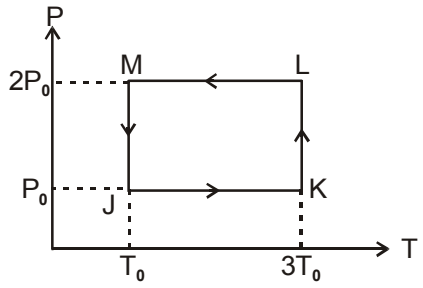
The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is : **[JEE Main Online 2024]**



(A) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^2$ (B) $\frac{V_a}{V_d}=\frac{V_b}{V_c}$

(C) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^{-1}$ (D) $\frac{V_a}{V_d}\neq\frac{V_b}{V_c}$

44. One mole of a monatomic ideal gas undergoes the cyclic process $J \to K \to L \to M \to J$, as shown in the P-T diagram.



Match the quantities mentioned in List-I with their values in List-II and choose the correct option.
[R is the gas constant] **[JEE Advanced 2024]**

	List-I		**List-II**
(P)	Work done in the complete cyclic process	(1)	$RT_0 - 4RT_0 \ln 2$
(Q)	Change in the internal energy of the gas in the process JK	(2)	0
(R)	Heat given to the gas in the process KL	(3)	$3RT_0$
(S)	Change in the internal energy of the gas in the process MJ	(4)	$-2RT_0 \ln 2$
		(5)	$-3RT_0 \ln 2$

(A) $P \to 1$; $Q \to 3$; $R \to 5$; $S \to 4$ (B) $P \to 4$; $Q \to 3$; $R \to 5$; $S \to 2$

(C) $P \to 4$; $Q \to 1$; $R \to 2$; $S \to 2$ (D) $P \to 2$; $Q \to 5$; $R \to 3$; $S \to 4$

45. Pressure of an ideal gas, contained in a closed vessel, is increased by 0.4% when heated by 1°C. Its initial temperature must be : **[JEE Main Online 2025]**

(A) 25°C (B) 2500 K (C) 250 K (D) 250°C

46. There are two vessels filled with an ideal gas where volume of one is double the volume of other. The large vessel contains the gas at 8 kPa at 1000 K while the smaller vessel contains the gas at 7 kPa at 500 K. If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K, at steady state the pressure in the vessels will be (in kPa).

[JEE Main Online 2025]

(A) 4.4 (B) 6 (C) 24 (D) 18

47. The mean free path and the average speed of oxygen molecules at 300 K and 1 atm are 3×10^{-7} m and 600 m/s, respectively. Find the frequency of its collisions. **[JEE Main Online 2025]**

(A) 2×10^{10}/s (B) 9×10^{5}/s (C) 2×10^{9}/s (D) 5×10^{8}/s

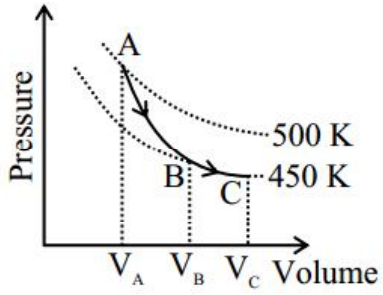
48. The helium and argon are put in the flask at the same room temperature (300 K). The ratio of average kinetic energies (per molecule) of helium and argon is :
(Give : Molar mass of helium = 4 g/mol, Molar mass of argon = 40 g/ mol)**[JEE Main Online 2025]**

(A) 1 : 10 (B) 10 : 1 (C) 1 : $\sqrt{10}$ (D) 1 : 1

49. For a diatomic gas, if $\gamma_1 = \left(\frac{Cp}{Cv}\right)$ for rigid molecules and $\gamma_2 = \left(\frac{Cp}{Cv}\right)$ for another diatomic molecules, but also having vibrational modes. Then, which one of the following options is correct ?
(Cp and Cv are specific heats of the gas at constant pressure and volume)**[JEE Main Online 2025]**

(A) $\gamma_2 > \gamma_1$ (B) $\gamma_2 = \gamma_1$ (C) $2\gamma_2 > \gamma_1$ (D) $\gamma_2 < \gamma_1$

50. A poly-atomic molecule (C_V = 3R, C_P = 4R, where R is gas constant) goes from phase space point A($P_A = 10^5$ Pa, $V_A = 4 \times 10^{-6}m^3$) to point B($P_B = 5 \times 10^4$ Pa, $V_B = 6 \times 10^{-6}m^3$) to point C($P_C = 10^4$ Pa, $V_C = 8 \times 10^{-6}m^3$). A to B is an adiabatic path and B to C is an isothermal path.
The net heat absorbed per unit mole by the system is : **[JEE Main Online 2025]**



(A) 500R(ln3 + ln4) (B) 450R(ln4 – ln3)
(C) 500R ln2 (D) 400R ln4

51. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : With the increase in the pressure of an ideal gas, the volume falls off more rapidly in an isothermal process in comparison to the adiabatic process.
Reason (R) : In isothermal process, PV = constant, while in adiabatic process PV^γ = constant. Here γ is the ratio of specific heats, P is the pressure and V is the volume of the ideal gas.
In the light of the above statements, choose the **correct** answer from the options given below :
[JEE Main Online 2025]

(A) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)
(B) (A) is true but (R) is false
(C) Both (A) and (R) are true and (R) is the correct explanation of (A).
(D) (A) is false but (R) is true

52. The workdone in an adiabatic change in an ideal gas depends upon only :**[JEE Main Online 2025]**
(A) change in its pressure (B) change in its specific heat
(C) change in its volume (D) change in its temperature

53. The ratio of vapour densities of two gases at the same temperature is $\frac{4}{25}$, then the ratio of r.m.s. velocities will be **[JEE Main Online 2025]**

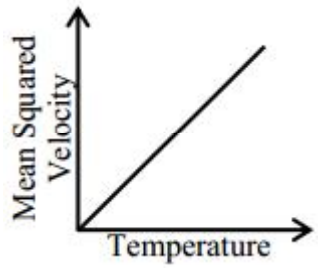
(A) $\frac{25}{4}$ (B) $\frac{2}{5}$ (C) $\frac{5}{2}$ (D) $\frac{4}{25}$

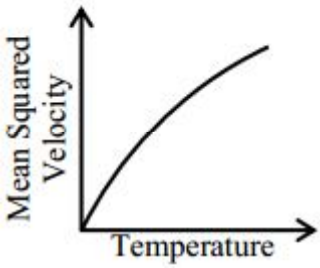
54. The kinetic energy of translation of the molecules in 50 g of CO_2 gas at 17°C is :

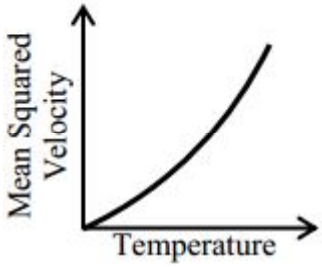
[JEE Main Online 2025]

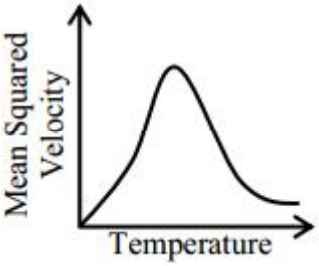
(A) 3986.3 J (B) 4102.8 J (C) 4205.5 J (D) 3582.7 J

55. For a particular ideal gas which of the following graphs represents the variation of mean squared velocity of the gas molecules with temperature ? **[JEE Main Online 2025]**

(A) 

(B) 

(C) 

(D) 

56. A Carnot engine (E) is working between two temperatures 473K and 273K. In a new system two engines - engine E_1 works between 473K to 373K and engine E_2 works between 373K to 273K. If η_{12}, η_1 and η_2 are the efficiencies of the engines E, E_1 and E_2, respectively, then **[JEE Main Online 2025]**

(A) $\eta_{12} < \eta_1 + \eta_2$ (B) $\eta_{12} = \eta_1 \eta_2$ (C) $\eta_{12} = \eta_1 + \eta_2$ (D) $\eta_{12} \geq \eta_1 + \eta_2$

57. Given below are two statements. One is labelled as **Assertion (A)** and the other is labelled as **Reason(R)**.

Assertion (A) : In an insulated container, a gas is adiabatically shrunk to half of its initial volume. The temperature of the gas decreases.

Reason (R) : Free expansion of an ideal gas is an irreversible and an adiabatic process.

In the light of the above statement, choose the **correct** answer from the options given below :

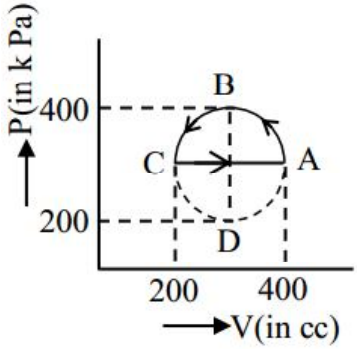
(A) Both (A) and (R) are true and (R) is the correct explanation of (A) **[JEE Main Online 2025]**

(B) (A) is true but (R) is false

(C) (A) is false but (R) is true

(D) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

58. The magnitude of heat exchanged by a system for the given cyclic process ABCA (as shown in figure) is (in SI unit) **[JEE Main Online 2025]**



(A) 10π (B) 5π (C) zero (D) 40π

59. Match the **List-I** with **List-II**

List-I	**List-II**
A. Pressure varies inversely with volume of an ideal gas.	I. Adiabatic process
B. Heat absorbed goes partly to increase internal energy and partly to do work.	II. Isochoric process
C. Heat is neither absorbed nor released by a system	III. Isothermal process
D. No work is done on or by a gas	IV. Isobaric process

Choose the correct answer from the options given below : **[JEE Main Online 2025]**

(A) A–I, B–IV, C–II, D–III (B) A–III, B–I, C–IV, D–II

(C) A–I, B–III, C–II, D–IV (D) A–III, B–IV, C–I, D–II

60. Given are statements for certain thermodynamic variables,

(I) Internal energy, volume (V) and mass (M) are extensive variables.

(II) Pressure (P), temperature (T) and density (ρ) are intensive variables.

(III) Volume (V), temperature (T) and density (ρ) are intensive variables.

(IV) Mass (M), temperature (T) and internal energy are extensive variables.

Choose the correct answer from the points given below : **[JEE Main Online 2025]**

(A) (III) and (IV) only (B) (IV) and (I) only (C) (I) and (II) only (D) (II) and (III) only

61. Water of mass m gram is slowly heated to increase the temperature from T_1 to T_2. The change in entropy of the water, given specific heat of water is 1 $Jkg^{-1}K^{-1}$, is : **[JEE Main Online 2025]**

(A) Zero (B) $m(T_2 - T_1)$ (C) $m\ell n\left(\frac{T_1}{T_2}\right)$ (D) $m\ell n\left(\frac{T_2}{T_1}\right)$

62. An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.

(I) The work done by gas during the process is zero.

(II) The heat added to gas is different from change in its internal energy.

(III) The volume of the gas is increased.

(IV) The internal energy of the gas is increased.

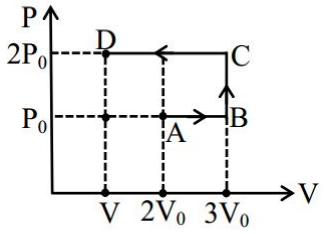
(V) The process is isochoric (constant volume process)

Choose the correct answer from the options given below **[JEE Main Online 2025]**

(A) (I), (II), (III), (IV) Only (B) (I), (IV), (V) Only

(C) (V) Only (D) (I), (III) Only

63. Using the given P-V diagram, the work done by an ideal gas along the path ABCD is **[JEE Main Online 2025]**



(A) $4\ P_0V_0$ (B) $3\ P_0V_0$ (C) $-4\ P_0V_0$ (D) $-3\ P_0V_0$

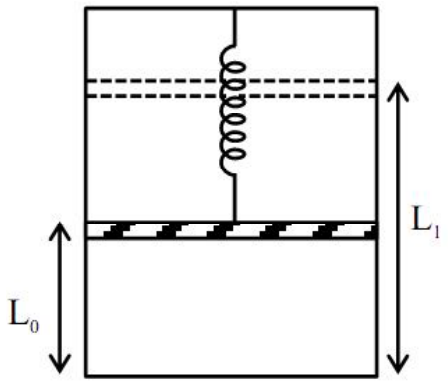
64. Identify the characteristics of an adiabatic process in a monoatomic gas.
(I) Internal energy is constant.
(II) Work done in the process is equal to the charge in internal energy.
(III) The product of temperature and volume is a constant.
(IV) The product of pressure and volume is a constant.
(V) The work done to change the temperature from T_1 to T_2 is proportional to $(T_2 - T_1)$
Choose the correct answer from the options given below : **[JEE Main Online 2025]**
(A) (I), (III), (IV) only (B) (I), (III), (V) only (C) (II), (V) only (D) (II), (IV) only

65. In an adiabatic process, which of the following statements is true? **[JEE Main Online 2025]**
(A) The molar heat capacity is infinite
(B) Work done by the gas equals the increase in internal energy
(C) The molar heat capacity is zero
(D) The internal energy of the gas decreases as the temperature increases

66. An ideal gas exists in a state with pressure P_0, volume V_0. It is isothermally expanded to 4 times of its initial volume (V_0), then isobarically compressed to its original volume. Finally the system is heated isochorically to bring it to its initial state. The amount of heat exchanged in this process is : **[JEE Main Online 2025]**
(A) $P_0V_0(2\ln 2 - 0.75)$ (B) $P_0V_0(\ln 2 - 0.75)$ (C) $P_0V_0(\ln 2 - 0.25)$ (D) $P_0V_0(2\ln 2 - 0.25)$

67. During the melting of a slab of ice at 273 K at atmospheric pressure : **[JEE Main Online 2025]**
(A) Internal energy of ice-water system remains unchanged.
(B) Positive work is done by the ice-water system on the atmosphere.
(C) Internal energy of the ice-water system decreases.
(D) Positive work is done on the ice-water system by the atmosphere.

68. A piston of mass M is hung from a massless spring whose restoring force law goes as $F = -kx^3$, where k is the spring constant of appropriate dimension. The piston separates the vertical chamber into two parts, where the bottom part is filled with 'n' moles of an ideal gas. An external work is done on the gas isothermally (at a constant temperature T) with the help of a heating filament (with negligible volume) mounted in lower part of the chamber, so that the piston goes up from a height L_0 to L_1, the total energy delivered by the filament is (Assume spring to be in its natural length before heating) **[JEE Main Online 2025]**



(A) $3nRT\ln\left(\frac{L_1}{L_0}\right) + 2Mg(L_1 - L_0) + \frac{k}{3}(L_1^3 - L_0^3)$ (B) $nRT\ln\left(\frac{L_1^2}{L_0}\right) + \frac{Mg}{2}(L_1 - L_0) + \frac{k}{4}(L_1^4 - L_0^4)$

(C) $nRT\ln\left(\frac{L_1}{L_0}\right) + Mg(L_1 - L_0) + \frac{k}{4}(L_1^4 - L_0^4)$ (D) $nRT\ln\left(\frac{L_1}{L_0}\right) + Mg(L_1 - L_0) + \frac{3k}{4}(L_1^4 - L_0^4)$

69. A gas is kept in a container having walls which are thermally non-conducting. Initially the gas has a volume of 800 cm^3 and temperature 27°C. The change in temperature when the gas is adiabatically compressed to 200 cm^3 is :
(Take $\gamma = 1.5$: γ is the ratio of specific heats at constant pressure and at constant volume)

[JEE Main Online 2025]

(A) 327 K (B) 600 K (C) 522 K (D) 300 K

70. Match **List-I** with **List-II**.

List-I	**List-II**
(a) Isobaric	(I) $\Delta Q = \Delta W$
(b) Isochoric	(II) $\Delta Q = \Delta U$
(c) Adiabatic	(III) ΔQ = zero
(d) Isothermal	(IV) $\Delta Q = \Delta U + P\Delta V$

ΔQ = Heat supplied
ΔW = Work done by the system
ΔU = Change in internal energy
P = Pressure of the system
ΔV = Change in volume of the system

Choose the correct answer from the options given below : **[JEE Main Online 2025]**

(A) (a)-(IV), (b)-(III), (c)-(II), (d)-(I) (B) (a)-(IV), (b)-(I), (c)-(III), (d)-(II)
(C) (a)-(IV), (b)-(II), (c)-(III), (d)-(I) (D) (a)-(II), (b)-(IV), (c)-(III), (d)-(I)

71. Match **List-I** with **List-II**.

List-I	**List-II**
(a) Isothermal	(I) ΔW (work done) = 0
(b) Adiabatic	(II) ΔQ (supplied heat) = 0
(c) Isobaric	(III) ΔU (change in internal energy) $\neq 0$
(d) Isochoric	(IV) $\Delta U = 0$

Choose the correct answer from the options given below : **[JEE Main Online 2025]**

(A) (a)-(III), (b)-(II), (c)-(I), (d)-(IV) (B) (a)-(IV), (b)-(I), (c)-(III), (d)-(II)
(C) (a)-(IV), (b)-(II), (c)-(III), (d)-(I) (D) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)

72. Match the **List-I** with **List-II**

List-I	**List-II**
A. Triatomic rigid gas	I. $\frac{C_P}{C_V} = \frac{5}{3}$
B. Diatomic non-rigid gas	II. $\frac{C_P}{C_V} = \frac{7}{5}$
C. Monoatomic gas	III. $\frac{C_P}{C_V} = \frac{4}{3}$
D. Diatomic rigid gas	IV. $\frac{C_P}{C_V} = \frac{9}{7}$

Choose the correct answer from the options given below : **[JEE Main Online 2025]**

(A) A-III, B-IV, C-I, D-II (B) A-III, B-II, C-IV, D-I (C) A-II, B-IV, C-I, D-III (D) A-IV, B-II, C-III, D-I

73. A monoatomic gas having $\gamma = \frac{5}{3}$ is stored in a thermally insulated container and the gas is suddenly compressed to $\left(\frac{1}{8}\right)^{th}$ of its initial volume. The ratio of final pressure and initial pressure is:

(λ is the ratio of specific heats of the gas at constant pressure and at constant volume)

[JEE Main Online 2025]

(A) 16 (B) 40 (C) 32 (D) 28

74. γ_A is the specific heat ratio of monoatomic gas A having 3 translational degrees of freedom. γ_B is the specific heat ratio of polyatomic gas B having 3 translational, 3 rotational degrees of freedom and 1 vibrational mode.

If $\frac{\gamma_A}{\gamma_B} = \left(1+\frac{1}{n}\right)$, then the value of n is _____. **[JEE Main Online 2025]**

75. The internal energy of air in 4 m × 4 m × 3 m sized room at 1 atmospheric pressure will be ___ × 10^6 J. (Consider air as diatomic molecule) **[JEE Main Online 2025]**

76. An ideal gas initially at 0ºC temperature, is compressed suddenly to one fourth of its volume. If the ratio of specific heat at constant pressure to that at constant volume is 3/2, the change in temperature due to the thermodynamics process is____K. **[JEE Main Online 2025]**

77. The temperature of 1 mole of an ideal monoatomic gas is increased by 50°C at constant pressure. The total heat added and change in internal energy are E_1 and E_2, respectively. If $\frac{E_1}{E_2} = \frac{x}{9}$ then the value of x is _____. **[JEE Main Online 2025]**

78. A container of fixed volume contains a gas at 27°C. To double the pressure of the gas, the temperature of gas should be raised to _____ °C. **[JEE Main Online 2025]**

79. The efficiency of a Carnot engine operating with a hot reservoir kept at a temperature of 1000 K is 0.4. It extracts 150 J of heat per cycle from the hot reservoir. The work extracted from this engine is being fully used to run a heat pump which has a coefficient of performance 10. The hot reservoir of the heat pump is at a temperature of 300 K. Which of the following statements is/are correct : **[JEE Advanced 2025]**

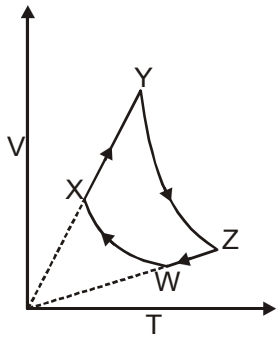
(A) Work extracted from the Carnot engine in one cycle is 60 J.

(B) Temperature of the cold reservoir of the Carnot engine is 600 K.

(C) Temperature of the cold reservoir of the heat pump is 270 K.

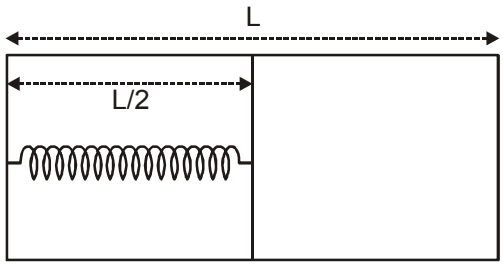
(D) Heat supplied to the hot reservoir of the heat pump in one cycle is 540 J.

80. An ideal monatomic gas of n moles is taken through a cycle WXYZW consisting of consecutive adiabatic and isobaric quasi-static processes, as shown in the schematic V-T diagram. The volume of the gas at W, X and Y points are, 64 cm^3, 125 cm^3 and 250 cm^3, respectively. If the absolute temperature of the gas T_W at the point W is such that nRT_W = 1 J (R is the universal gas constant), then the amount of heat absorbed (in J) by the gas along the path XY is _____.



[JEE Advanced 2025]

81. The left and right compartments of a thermally isolated container of length L are separated by a thermally conducting, movable piston of area A. The left and right compartments are filled with $\frac{3}{2}$ and 1 moles of an ideal gas, respectively. In the left compartment the piston is attached by a spring with spring constant k and natural length $\frac{2L}{5}$. In thermodynamic equilibrium, the piston is at a distance $\frac{L}{2}$ from the left and right edges of the container as shown in the figure. Under the above conditions, if the pressure in the right compartment is $P = \frac{\kappa L}{A}\alpha$, then the value of α is _____. **[JEE Advanced 2025]**



ANSWER KEY

EXERCISE-I

1.	A	**2.**	A	**3.**	B	**4.**	A	**5.**	B	**6.**	A	**7.**	C
8.	C	**9.**	D	**10.**	B	**11.**	A	**12.**	B	**13.**	C	**14.**	D
15.	A	**16.**	C	**17.**	B	**18.**	D	**19.**	C	**20.**	A	**21.**	D
22.	D	**23.**	A	**24.**	C	**25.**	D	**26.**	D	**27.**	C	**28.**	C
29.	D	**30.**	A	**31.**	A	**32.**	A	**33.**	A	**34.**	D	**35.**	D
36.	A	**37.**	D	**38.**	C	**39.**	D	**40.**	A	**41.**	D	**42.**	C
43.	D	**44.**	A	**45.**	A	**46.**	C	**47.**	B	**48.**	D	**49.**	C
50.	C	**51.**	D	**52.**	B	**53.**	A	**55.**	B	**56.**	C	**57.**	C
58.	D	**59.**	BD	**60.**	C	**61.**	AD	**62.**	D	**63.**	C	**64.**	AC
65.	D	**66.**	ABC	**67.**	C	**68.**	B	**69.**	BC	**70.**	AC	**71.**	D
72.	AB	**73.**	$A \to P,Q$; $B \to RS$; $C \to PQ$					**74.**	$A \to Q$; $B \to R$; $C \to P$; $D \to Q$				

EXERCISE - II

1. 327 °C **2.** 4/3 **3.** 4 : 1

4. (i) $P_1 < P_2$, $T_1 < T_2$; (ii) $T_1 = T_2 < T_3$; (iii) $V_2 > V_1$; (iv) $P_1 > P_2$

5. 1.84P, 10V/3, 8V/3 (adiabatic), 13P/6, 30V/13, 48V/13 (isothermal)

6. $\frac{3}{2}\rho_0$ **7.** 100 J **8.** 3600 R **9.** 400 K **10.** 12600 J **11.** 50 calorie

12. $R\Delta T\left(\frac{3-2\gamma}{\gamma-1}\right)$ **13.** 1.5 **14.** $\frac{mRT_0}{V_0}\left(1+\frac{T_0 m}{V_0}\right)R$ **15.** $RT\left[1-\frac{1}{n}\right]$

16. P/n **17.** 5R **18.** $\frac{R}{2}$ **19.** (i) $\left(\frac{\gamma+1}{\gamma-1}\right) 4aV_0^2$, (ii) $\left(\frac{\gamma+1}{\gamma-1}\right)\frac{R}{2}$

20. (i) 74 cm, (ii) 73.94 cm, (iii) 69.52 cm

21. $p_1 = p_{H_2} \simeq 1.25 \times 10^6$ Pa ; $p_2 = p_{H_2} + p_{O_2} + p_{N_2} \simeq 2.8125 \times 10^6$ Pa ; $p_3 = p_{H_2} + p_{N_2} \simeq 1.5625 \times 10^6$ Pa

22. (ii) $P_b = P_c = 2$ atm, (iii) $T_b = 300$ K, $T_c = 600$ K, (iv) $V_c = 8$ litre **23.** 1.6 m, 364 K

24. 1.63 **25.** $2mnv^2\cos^2\theta$ **26.** C = 2 R **27.** $Q_1 < Q_2$

EXERCISE - III

1.	ABD	**2.**	C	**3.**	BCD	**4.**	C	**5.**	900
6.	BCD	**7.**	C	**8.**	A	**9.**	B	**10.**	C
11.	0.29 - 0.31	**12.**	9.00 - 10.00	**13.**	A	**14.**	B	**15.**	B
16.	B	**17.**	C	**18.**	B	**19.**	D	**20.**	C
21.	250	**22.**	7479	**23.**	12	**24.**	D	**25.**	C
26.	C	**27.**	B	**28.**	C	**29.**	D	**30.**	D
31.	B	**32.**	C	**33.**	A	**34.**	2.00	**35.**	121.0
36.	D	**37.**	C	**38.**	B	**39.**	B	**40.**	B
41.	B	**42.**	A	**43.**	B	**44.**	B	**45.**	C
46.	B	**47.**	C	**48.**	D	**49.**	D	**50.**	B
51.	D	**52.**	D	**53.**	C	**54.**	B	**55.**	A
56.	A	**57.**	C	**58.**	B	**59.**	D	**60.**	C
61.	D	**62.**	B	**63.**	D	**64.**	C	**65.**	C
66.	A	**67.**	D	**68.**	C	**69.**	D	**70.**	C
71.	C	**72.**	A	**73.**	C	**74.**	03.00	**75.**	12.00
76.	273	**77.**	15.00	**78.**	327	**79.**	A, B, C	**80.**	1.60
81.	0.20								